

## Lecture 8

# Lax pairs, and the spectral parameter

**Objectives.** Course map: G1, step 8 of 9. [Lecture 5](#) said what  $n$  integrals in involution buy, and nothing about how to find them. This lecture gives the method that finds them for the Toda chain: write the equations of motion as a matrix whose spectrum does not change. We prove that the spectrum is conserved, derive the Toda chain's Lax matrix, close the chain into a ring, and rewrite the ring's eigenvalue problem as a  $2 \times 2$  site matrix with a spectral parameter. We also say plainly what the method does not give: that the integrals commute. [Lecture 9](#) proves that, from the site matrix.

## 8.1 The question: where do the integrals come from?

[Definition 5.5](#) asks for  $n$  independent first integrals in involution, and [Theorem 5.7](#) turns them into tori. Neither says where the integrals come from. The ones we own came from two sources: the energy, and symmetry — angular momentum for a central force, momentum for a translation invariant system.

That is not enough for the models this course is built around. The open Toda chain of  $N$  particles,

$$H = \frac{1}{2} \sum_{n=1}^N p_n^2 + \sum_{n=1}^{N-1} e^{q_n - q_{n+1}}, \quad (8.1)$$

has two obvious integrals,  $H$  and the total momentum  $\sum_n p_n$ , and no other evident continuous symmetry: apart from a common translation, no one-parameter family of transformations of the positions leaves  $H$  invariant. In the early 1970s Ford's numerical experiments on the periodic chain suggested that it was nevertheless integrable, and in 1974 Hénon and Flaschka independently proved it (Toda, §2.10). Flaschka's construction, done here for the open chain, is the subject of § 8.4. It works by finding a matrix  $L$ , built from the  $q$ 's and  $p$ 's, whose eigenvalues do not move.  $N$  eigenvalues,  $N$  integrals.

So the question of this lecture is

*where do the missing integrals of a many-body chain come from?*

The answer is a Lax pair. We derive the Toda chain's, close the chain into a ring, and rewrite the ring's eigenvalue problem as a  $2 \times 2$  transport from site to site, with a spectral parameter (§ 8.7).

## 8.2 The Lax equation and isospectrality

**Definition 8.1 (Lax pair).** Let  $H$  be a Hamiltonian on a phase space  $\mathcal{M}$ . A Lax pair for  $H$  is a pair of smooth functions  $L, M: \mathcal{M} \rightarrow \text{Mat}_N(\mathbb{C})$  such that along every solution  $x(t)$  of Hamilton's equations,  $L(t) = L(x(t))$  and  $M(t) = M(x(t))$  satisfy the Lax equation

$$\frac{dL}{dt} = [M, L] = ML - LM. \quad (8.2)$$

$L$  is the Lax matrix;  $N$  is its size, unrelated to  $\dim \mathcal{M}$ .

The definition asks that Hamilton's equations *imply* (8.2). In good cases the converse holds too, and (8.2) is the equations of motion, written as a matrix identity; we will see this for Toda, up to the motion of the centre of mass, which the Lax matrix cannot see.

**Theorem 8.2 (Isospectrality).** Let  $(L, M)$  be a Lax pair for  $H$  and  $x(t)$  a solution of Hamilton's equations defined on an interval containing 0. Then:

- (i)  $\text{tr } L^k$  is constant in  $t$  for every  $k \geq 1$ ;
- (ii)  $L(t) = U(t) L(0) U(t)^{-1}$ , where  $U$  solves  $\dot{U} = M(t)U$ ,  $U(0) = \mathbb{1}$ , and is invertible;
- (iii) the characteristic polynomial  $\det(\eta - L(t))$  is independent of  $t$ : the eigenvalues of  $L$ , with their multiplicities, are conserved.

Consequently each  $I_k = \text{tr } L^k$  is a first integral,  $\{I_k, H\} = 0$  (Definition 2.3); if  $L$  is complex, the real and imaginary parts of each  $I_k$  are. If  $M$  is antisymmetric,  $U(t)$  is orthogonal; if  $M$  is anti-Hermitian, unitary.

*Proof.* (i) By the product rule and (8.2), and then because the trace is invariant under cyclic permutation,

$$\frac{d}{dt} \text{tr } L^k = k \text{tr}(L^{k-1} \dot{L}) = k \text{tr}(L^{k-1} ML - L^k M) = 0.$$

(The product rule gives  $k$  terms  $\text{tr}(L^j \dot{L} L^{k-1-j})$ ; cyclicity makes them equal.)

(ii) The equation  $\dot{U} = M(t)U$  is linear with coefficients continuous in  $t$ , so its solution exists on the whole interval.<sup>1</sup> Let  $V$  solve  $\dot{V} = -V M(t)$ ,  $V(0) = \mathbb{1}$ . Then  $\frac{d}{dt}(VU) = -VMU + VMU = 0$ , so  $VU = \mathbb{1}$  and  $U$  is invertible with  $U^{-1} = V$ . Now

$$\frac{d}{dt}(VLU) = -VMLU + V(ML - LM)U + VLMU = 0,$$

so  $U^{-1}L(t)U = L(0)$ .

(iii)  $\det(\eta - ULU^{-1}) = \det(U(\eta - L)U^{-1}) = \det(\eta - L)$ .

Every point of  $\mathcal{M}$  lies on some solution, and  $dI_k/dt = \{I_k, H\}$  (Definition 2.3), so (i) gives  $\{I_k, H\} = 0$  everywhere. Finally, if  $M^T = -M$  then  $\frac{d}{dt}(U^T U) = U^T(M^T + M)U = 0$ , and the same line with the conjugate transpose  $U^*$  gives the unitary case. ■

Part (ii) says more than (i): the flow moves  $L$  *inside one conjugacy class*. For  $L$  real symmetric and  $M$  antisymmetric,  $[M, L]$  is symmetric, so  $L(t)$  stays symmetric and moves by orthogonal conjugation (Figure 8.1).

For an  $N \times N$  matrix only  $N$  of the  $\text{tr } L^k$  can be independent: by the Cayley–Hamilton theorem  $L^N$  is a combination of  $\mathbb{1}, L, \dots, L^{N-1}$  with coefficients that are polynomials in  $\text{tr } L, \dots, \text{tr } L^N$ , so every  $\text{tr } L^k$  with  $k > N$  is a polynomial in the first  $N$ . Equivalently, the conserved data is the characteristic polynomial, which has  $N$  coefficients.

<sup>1</sup>A linear equation cannot blow up in finite time: Grönwall's inequality bounds  $\|U(t)\|$  by  $e^{\int \|M\|}$ . This is the only fact about ODEs used beyond Theorem 1.5.

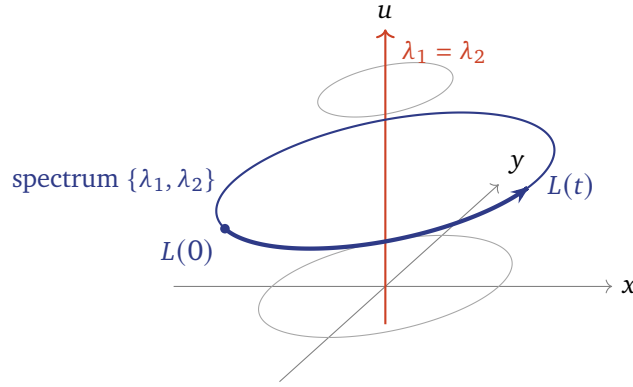


Figure 8.1: Isospectral flow in the smallest case: the real symmetric matrices  $\begin{pmatrix} u+x & y \\ y & u-x \end{pmatrix}$  with a given spectrum  $u \pm \sqrt{x^2 + y^2}$  form one horizontal circle, which a Lax flow with antisymmetric  $M$  cannot leave.

### 8.3 What a Lax pair does not give

[Theorem 8.2](#) produces first integrals, and [Definition 5.5](#) needs  $n$  of them, independent and in involution. These are three different things. *Having* a Lax pair is no property at all:  $L = (H)$  and  $M = (0)$  form one for every  $H$ . What matters is the spectral invariants a Lax pair delivers, and nothing in (8.2) makes them commute or makes enough of them independent. The Lax equation says how  $L$  moves along one flow, the flow of  $H$ ; involution is about the brackets of the  $\text{tr } L^k$  with *each other*, which no single flow can see. For a particle in a central potential,  $L = \text{diag}(\ell_1, \ell_2)$  and  $M = 0$  form a Lax pair, and by (5.3) and Leibniz

$$\{\text{tr } L, \text{tr } L^2\} = \{\ell_1 + \ell_2, \ell_1^2 + \ell_2^2\} = 2\ell_3(\ell_2 - \ell_1) \neq 0.$$

[Lecture 9](#) pays this debt for the Toda ring: it proves involution from a condition on the brackets of the entries of a Lax matrix, and checks independence.

### 8.4 The open Toda chain: Flaschka variables, $L$ and $M$ derived

$N$  unit masses on a line, positions  $q_1, \dots, q_N$ , each repelled by its neighbours through the potential  $e^{q_n - q_{n+1}}$ , which is large when particle  $n$  has overtaken particle  $n+1$  and small when they are far apart. With  $H$  as in (8.1), Hamilton's equations are

$$\dot{q}_n = p_n, \quad \dot{p}_n = e^{q_{n-1} - q_n} - e^{q_n - q_{n+1}}, \quad (8.3)$$

where the terms involving  $q_0$  and  $q_{N+1}$  are absent: the chain is *open*, and its end particles have one neighbour each. [Lecture 0](#) met this potential. Here we find its Lax pair in two steps, after a change of variables that makes the equations polynomial.

#### Flaschka's variables

The exponentials are the obstacle. Set

$$a_n = e^{(q_n - q_{n+1})/2} \quad (n = 1, \dots, N-1), \quad b_n = p_n \quad (n = 1, \dots, N). \quad (8.4)$$

Then  $a_n^2 = e^{q_n - q_{n+1}}$ , and (8.3) becomes

$$\dot{a}_n = \frac{1}{2} a_n (b_n - b_{n+1}), \quad \dot{b}_n = a_{n-1}^2 - a_n^2, \quad (8.5)$$

with the convention  $a_0 = a_N = 0$  — quadratic, with no exponential left. The map  $(q, p) \mapsto (a, b)$  forgets one thing: shifting every  $q_n$  by the same constant changes no  $a_n$ . The centre of mass position is invisible to  $L$ ; it is recovered from  $\dot{q}_n = p_n$  once the rest is known. Note also  $a_n > 0$ .

### Step 1: $L$

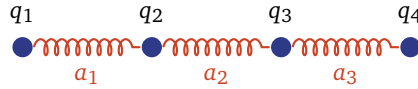
In the new variables

$$2H = \sum_{n=1}^N b_n^2 + 2 \sum_{n=1}^{N-1} a_n^2,$$

a sum of squares, each  $a_n^2$  counted twice. For a real symmetric matrix,  $\text{tr } L^2 = \sum_{j,k} L_{jk}^2$  is also a sum of squares, each off-diagonal entry counted twice because it appears as  $L_{jk}$  and  $L_{kj}$ . So the simplest  $L$  with  $\text{tr } L^2 = 2H$  has each  $b_n$  alone in a diagonal slot and each  $a_n$  alone in a symmetric pair of off-diagonal slots. Which pair is a choice, and locality makes it:  $a_n$  is the bond between particles  $n$  and  $n+1$ , so it goes in the slots  $(n, n+1)$  and  $(n+1, n)$ . The result is the symmetric tridiagonal (*Jacobi*) matrix of Figure 8.2; for  $N = 3$ ,

$$L = \begin{pmatrix} b_1 & a_1 & 0 \\ a_1 & b_2 & a_2 \\ 0 & a_2 & b_3 \end{pmatrix}, \quad \text{tr } L = \sum_n p_n, \quad \text{tr } L^2 = 2H. \quad (8.6)$$

Step 1 is a well-motivated choice, not a theorem; Step 2 is where it is tested.



$$L = \begin{array}{|c|c|c|c|} \hline b_1 & a_1 & 0 & 0 \\ \hline a_1 & b_2 & a_2 & 0 \\ \hline 0 & a_2 & b_3 & a_3 \\ \hline 0 & 0 & a_3 & b_4 \\ \hline \end{array}$$

bond  $n$  (red) fills the two slots  $(n, n+1)$  and  $(n+1, n)$ ;  
particle  $n$ 's momentum (indigo) fills  $(n, n)$ ; everything else is zero

Figure 8.2: Where the entries of Toda's  $L$  come from, for  $N = 4$ . The chain's nearest-neighbour structure becomes the tridiagonal band.

### Step 2: $M$ , from an ansatz

We want  $M$  with  $[M, L] = \dot{L}$ , where  $\dot{L}$  is given by (8.5). Take  $M$  antisymmetric, so that  $L$  stays symmetric.  $\dot{L}$  is tridiagonal, and the commutator of two tridiagonal matrices has bandwidth at most two, so try the least that could work:  $M$  antisymmetric and tridiagonal,

$$M_{n,n+1} = -m_n, \quad M_{n+1,n} = m_n,$$

with  $m_1, \dots, m_{N-1}$  unknown functions. Compare the three bands of  $[M, L]$  with those of  $\dot{L}$ . *Second off-diagonal.*  $\dot{L}$  has none, but

$$[M, L]_{n,n+2} = M_{n,n+1}L_{n+1,n+2} - L_{n,n+1}M_{n+1,n+2} = -(m_n a_{n+1} - a_n m_{n+1})$$

must vanish. So  $m_n/a_n = m_{n+1}/a_{n+1}$  for every  $n$ :  $m_n = c a_n$  with one common factor  $c$ .

*First off-diagonal.* Only two terms survive:

$$[M, L]_{n,n+1} = M_{n,n+1}L_{n+1,n+1} - L_{n,n}M_{n,n+1} = m_n(b_n - b_{n+1}),$$

and (8.5) asks for  $\frac{1}{2}a_n(b_n - b_{n+1})$ . As this must hold for all  $b$ ,  $m_n = \frac{1}{2}a_n$ :  $c = \frac{1}{2}$ , determined.

*Diagonal.* Nothing is left to choose, so this is the test:

$$[M, L]_{nn} = M_{n,n+1}a_n + M_{n,n-1}a_{n-1} - a_nM_{n+1,n} - a_{n-1}M_{n-1,n} = 2(m_{n-1}a_{n-1} - m_na_n) = a_{n-1}^2 - a_n^2,$$

which is  $\dot{b}_n$ . It could have failed; it does not.

**Proposition 8.3 (The Toda Lax pair).** *With  $L$  the symmetric tridiagonal matrix with diagonal  $b_n$  and off-diagonal  $a_n$ , and  $M$  the antisymmetric tridiagonal matrix with*

$$M_{n,n+1} = -\frac{1}{2}a_n, \quad M_{n+1,n} = \frac{1}{2}a_n, \quad (8.7)$$

*the Toda equations (8.5) are equivalent to  $\dot{L} = [M, L]$ . Among antisymmetric matrices,  $M$  is the only solution.*

*Proof.* The three band computations above prove the Lax equation, and show that the tridiagonal ansatz has exactly this solution. The equivalence holds because the bands of  $\dot{L} = [M, L]$  are (8.5), entry by entry. For uniqueness among all antisymmetric matrices: if  $M'$  also works,  $D = M - M'$  is antisymmetric and commutes with  $L$ . A Jacobi matrix with all  $a_n \neq 0$  has  $N$  distinct eigenvalues: if  $Lv = \eta v$ , row 1 gives  $v_2$  from  $v_1$ , row 2 gives  $v_3$ , and so on, so each eigenspace is at most one-dimensional. Diagonalise  $L = O\Lambda O^T$  with  $O$  orthogonal; then  $O^T D O$  commutes with the diagonal  $\Lambda$  of distinct entries, so it is diagonal, and it is antisymmetric, so it is zero. ■

So, to “where did  $L$  come from?”: from  $\text{tr } L^2 = 2H$  and locality. And “where did  $M$  come from?”: it was solved for, and it was the only possibility.

## The integrals

By Theorem 8.2, the  $N$  functions  $\text{tr } L^k$ ,  $k = 1, \dots, N$ , are first integrals. The first three:

$$\text{tr } L = \sum_n p_n, \quad \text{tr } L^2 = 2H, \quad \text{tr } L^3 = \sum_n p_n^3 + 3 \sum_{n=1}^{N-1} e^{q_n - q_{n+1}} (p_n + p_{n+1}). \quad (8.8)$$

Total momentum, energy, and an integral cubic in the momenta. By Lecture 2’s one-line Noether theorem it too generates a symmetry of  $H$ , but not a transformation of the positions alone, since its flow moves each  $q_n$  at a rate  $\partial(\text{tr } L^3)/\partial p_n$  that depends on the momenta. That is why nobody saw it. Figure 8.3 watches them at work for  $N = 3$ : the momenta change completely during a collision while the three eigenvalues of  $L$  do not change at all. What the eigenvalues mean is Lecture 12’s business, and the figure already hints at it.

## 8.5 Gauge freedom

A Lax pair is not unique, and the non-uniqueness has a structure worth knowing.

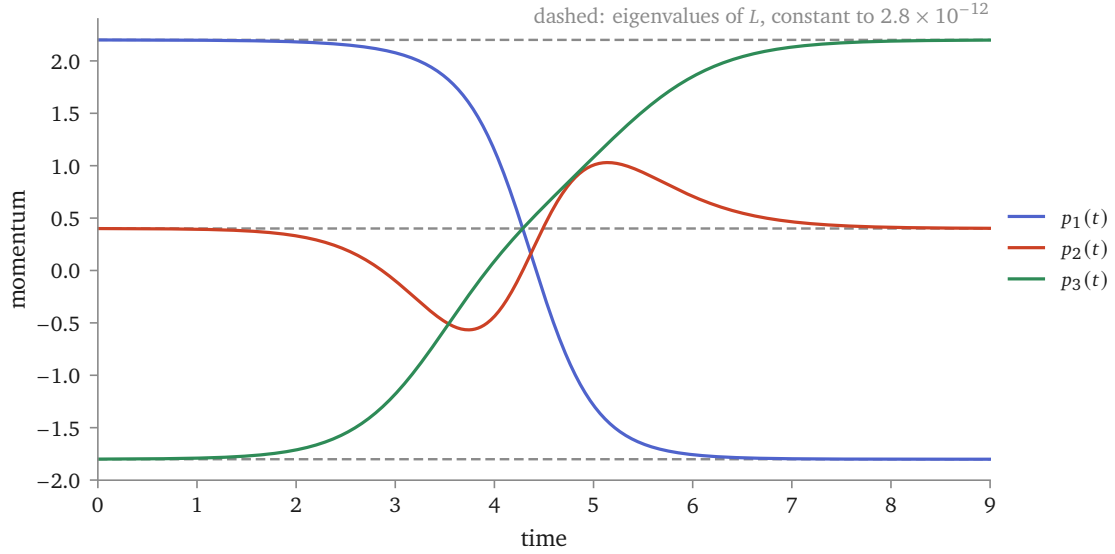


Figure 8.3: The open Toda chain,  $N = 3$ : momenta  $p_n(t)$  (solid) against the eigenvalues of  $L$  (dashed grey), which stay constant while the momenta are rearranged (scripts/figures/toda\_isospectral.py).

**Proposition 8.4** (Gauge transformations). *Let  $(L, M)$  satisfy (8.2) along a solution, and let  $g(t)$  be invertible and differentiable in  $t$  (for instance a function on phase space, evaluated along the solution). Then*

$$\tilde{L} = gLg^{-1}, \quad \tilde{M} = gMg^{-1} + \dot{g}g^{-1} \quad (8.9)$$

*satisfy  $\dot{\tilde{L}} = [\tilde{M}, \tilde{L}]$ , and  $\tilde{L}$  has the same spectrum as  $L$ .*

*Proof.* With  $\frac{d}{dt}g^{-1} = -g^{-1}\dot{g}g^{-1}$ ,

$$\dot{\tilde{L}} = \dot{g}Lg^{-1} + g[M, L]g^{-1} - gLg^{-1}\dot{g}g^{-1} = [\dot{g}g^{-1}, \tilde{L}] + [gMg^{-1}, \tilde{L}]. \quad \blacksquare$$

Everything spectral is gauge invariant. Everything else is not: symmetry, the band structure,  $M$  itself. For Toda, conjugate by  $D = \text{diag}(d_n)$  with  $d_1 = 1$ ,  $d_{n+1} = a_n d_n$ . Then  $DLD^{-1}$  has  $b_n$  on the diagonal, 1 above it and  $a_n^2 = e^{q_n - q_{n+1}}$  below it: not symmetric, but with entries that are simpler functions of  $(q, p)$ , and with the same characteristic polynomial. A reading at the end of the lecture draws the two as different paths on one isospectral set.

## 8.6 Periodic boundary conditions

Put  $N$  particles on a ring and let particle  $N$  interact with particle 1 across the join (Figure 8.4). With the positions read cyclically,  $q_{n+N} = q_n$ , the Hamiltonian is

$$H = \frac{1}{2} \sum_{n=1}^N p_n^2 + \sum_{n=1}^N e^{q_n - q_{n+1}}, \quad q_{N+1} = q_1, \quad (8.10)$$

the open chain (8.1) with one more spring,  $e^{q_N - q_1}$ . The equations of motion are those of the open chain with every index read modulo  $N$ :  $\dot{q}_n = p_n$ ,  $\dot{p}_n = e^{q_{n-1} - q_n} - e^{q_n - q_{n+1}}$ . For  $N = 3$ , in centre-of-mass coordinates and after a rescaling, this is the Hamiltonian (7.7) whose

Poincaré section [Figure 7.5](#) showed smooth curves where Hénon–Heiles had chaos. [Lecture 9](#) proves that what the figure showed is integrability.

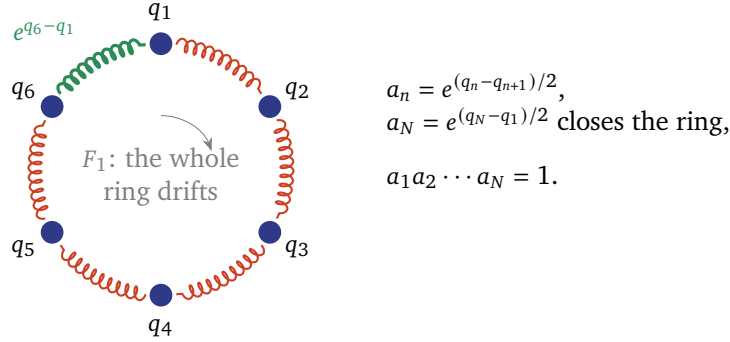


Figure 8.4: The periodic Toda chain,  $N = 6$ . The new spring (green) joins particle  $N$  to particle 1. A common translation changes no spring, so the flow of  $F_1 = \sum_n p_n$  carries the whole ring along without end.

### The Lax pair of the ring

Flaschka's variables (8.4) carry over with one more bond,  $a_N = e^{(q_N - q_1)/2}$ , and one new constraint: the exponents telescope to zero round the ring, so

$$a_1 a_2 \cdots a_N = 1. \quad (8.11)$$

Put  $a_N$  in the two corners:

$$L = \begin{pmatrix} b_1 & a_1 & & & a_N \\ a_1 & b_2 & a_2 & & \\ & \ddots & \ddots & \ddots & \\ & & a_{N-2} & b_{N-1} & a_{N-1} \\ a_N & & & a_{N-1} & b_N \end{pmatrix}, \quad (8.12)$$

Let  $M$  be antisymmetric with  $M_{n,n+1} = -\frac{1}{2}a_n$  for every  $n$ , indices modulo  $N$ , so that  $M_{N1} = -\frac{1}{2}a_N = -M_{1N}$ . The band computations of [Proposition 8.3](#) go through with the indices read modulo  $N$  (for  $N = 3$  the second-neighbour terms, which cancel identically, land on first-neighbour slots and change nothing), so  $\dot{L} = [M, L]$  for  $N \geq 3$ , and by [Theorem 8.2](#) the  $F_k = \text{tr } L^k$  are conserved, with  $F_1 = \sum_n p_n$  and  $F_2 = 2H$ .

## 8.7 The site matrix

The eigenvalue problem  $L\psi = \lambda\psi$  of the ring's Lax matrix is a recursion,

$$a_{n-1}\psi_{n-1} + b_n\psi_n + a_n\psi_{n+1} = \lambda\psi_n, \quad (8.13)$$

closed up by  $\psi_0 = \psi_N$  and  $\psi_{N+1} = \psi_1$ . The recursion (8.13) is of second order, so it is a first-order recursion for pairs  $(\psi_{n+1}, \psi_n)$ . Solving for  $\psi_{n+1}$  means dividing by  $a_n$ , which gives a  $2 \times 2$  matrix of determinant  $a_{n-1}/a_n$  involving two bonds. A gauge transformation does better.

### The gauge that makes it local

Conjugate by  $D = \text{diag}(e^{-q_1/2}, \dots, e^{-q_N/2})$ , the gauge of [Proposition 8.4](#) (it is § 8.5's  $D$  up to a constant). In the new variable  $\phi_n = e^{-q_n/2}\psi_n$ , (8.13) reads

$$\phi_{n+1} = (\lambda - p_n) \phi_n - e^{q_{n-1}-q_n} \phi_{n-1} : \quad (8.14)$$

the coefficient of  $\phi_{n+1}$  is now 1. (Substitute  $\psi_n = e^{q_n/2}\phi_n$  in (8.13), use  $a_n e^{q_{n+1}/2} = e^{q_n/2}$ , and divide by  $e^{q_n/2}$ .) Now collect a pair,  $\Phi_n = (\phi_n, -e^{q_{n-1}}\phi_{n-1})^T$ . Then (8.14) and the definition of the second component say exactly

$$\Phi_{n+1} = L_n(\lambda) \Phi_n, \quad L_n(\lambda) = \begin{pmatrix} \lambda - p_n & e^{-q_n} \\ -e^{q_n} & 0 \end{pmatrix}, \quad \det L_n(\lambda) = 1. \quad (8.15)$$

This is the *site matrix*: local (site  $n$  only), polynomial in the spectral parameter  $\lambda$ , and a transport one step along the lattice ([Figure 8.5](#)).

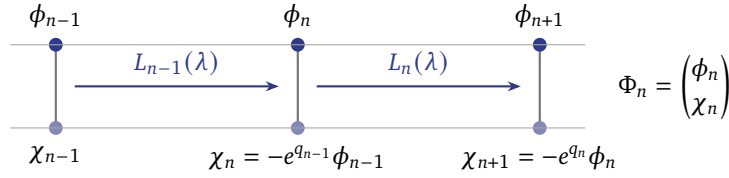


Figure 8.5: The recursion (8.14) as a ladder: the site matrix  $L_n(\lambda)$ , built from  $(q_n, p_n)$  alone, carries each rung  $\Phi_n$  to the next.

## 8.8 What it buys

- *N integrals for the Toda ring*, the  $\text{tr } L^k$  for  $k = 1, \dots, N$ , from one matrix whose spectrum does not move. Energy and total momentum are two of them; the rest were invisible as symmetries of the positions.
- *A site matrix with a spectral parameter*. The ring's eigenvalue problem is a product of  $2 \times 2$  matrices  $L_n(\lambda)$  (8.15), each built from one site's  $(q_n, p_n)$  alone. [Lecture 9](#) multiplies them round the ring and proves, from one identity that a single site matrix satisfies, that the integrals commute.
- *A method*: choose  $L$  from  $\text{tr } L^2 = 2H$  and locality, then solve for  $M$ .

## 8.9 Caveats, in one place

- *Status*. [Theorem 8.2](#), [Proposition 8.3](#) and [Proposition 8.4](#) are proved. The ring's Lax pair is the band computation of [Proposition 8.3](#) with indices read modulo  $N$ , and the site matrix (8.15) is derived.
- *Conserved is not in involution* (§ 8.3); [Lecture 9](#) proves involution, and independence, for the ring.
- *The ring still drifts*. The flow of  $F_1$ ,  $q_n \mapsto q_n + s$ , changes no  $a_n$ , so no level set is compact; [Lecture 9](#) splits the drift off.
- *$N = 2$  is degenerate*. The corner slot (1, 2) is the off-diagonal slot, so  $L_{12} = a_1 + a_2$  and  $\text{tr } L^2 = 2H + 4$ . The formulas here are for  $N \geq 3$ .

## 8.10 The Euler top: a Lax pair whose spectrum misses an integral

A rigid body turning freely about its centre of mass, described in the *body frame*, the principal axes of inertia moving with the body. The principal moments are  $I_1, I_2, I_3 > 0$ ; the angular momentum and the angular velocity have body components  $\ell$  and  $\omega$ , with  $\ell_i = I_i \omega_i$ . In space  $\ell$  is constant; the body frame turns with angular velocity  $\omega$ , so  $0 = \dot{\ell} + \omega \times \ell$ , which are *Euler's equations*

$$\dot{\ell} = \ell \times \omega, \quad \omega_i = \ell_i / I_i, \quad (8.16)$$

in components  $I_1 \dot{\omega}_1 = (I_2 - I_3) \omega_2 \omega_3$  and cyclically. We use them only as a differential equation on  $\mathbb{R}^3$ . (These  $I_i$  are moments of inertia, not the integrals  $I_k$  of [Theorem 8.2](#) nor [Lecture 6](#)'s actions; the standard names are kept.) (The phase space of the body is  $T^*\text{SO}(3)$ , of dimension 6; the three components of  $\ell$  are functions on it, and (8.16) is the part of the motion that does not involve the orientation.)

### The Lax form

For  $a \in \mathbb{R}^3$  let  $\hat{a}$  be the antisymmetric matrix with  $\hat{a}v = a \times v$ :

$$\hat{a} = \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix}.$$

Then  $[\hat{a}, \hat{b}] = \widehat{a \times b}$ : apply both sides to  $v$  and use the Jacobi identity of the cross product,  $a \times (b \times v) - b \times (a \times v) = (a \times b) \times v$ . Hence, with  $L = \hat{\ell}$  and  $\Omega = \hat{\omega}$ , Euler's equations are

$$\frac{dL}{dt} = \widehat{\ell \times \omega} = [L, \Omega], \quad (8.17)$$

a Lax equation with  $M = -\Omega$ . The student already knew it; only the notation is new.

What does it conserve?  $L$  is real antisymmetric, so  $\text{tr } L = \text{tr } L^3 = 0$  and  $\text{tr } L^2 = -2|\ell|^2$ ; indeed

$$\det(\eta - L) = \eta^3 + |\ell|^2 \eta, \quad \text{eigenvalues } 0, \pm i|\ell|. \quad (8.18)$$

The spectrum knows  $|\ell|^2$  and nothing else. But the energy

$$H = \sum_i \frac{\ell_i^2}{2I_i}$$

is also conserved, since  $\dot{H} = \omega \cdot \dot{\ell} = \omega \cdot (\ell \times \omega) = 0$ , and unless  $I_1 = I_2 = I_3$  it is not a function of  $|\ell|^2$ . This is the insufficiency discussed in [§ 8.3](#): a genuine Lax pair, for a genuinely integrable system, whose spectrum misses the second integral. [§ 8.11](#) gets it back.

### The motion, from the two integrals

Take  $I_1 < I_2 < I_3$ . Both integrals together confine  $\ell(t)$  to the intersection of the sphere  $|\ell| = \ell_0 > 0$  with the ellipsoid  $\sum \ell_i^2 / I_i = 2H$  ([Figure 8.6](#)). On the sphere  $H$  is smallest at  $\pm \ell_0 e_3$  and largest at  $\pm \ell_0 e_1$ , and near these four points the level curves are small closed loops.

The equilibria of (8.16), where  $\ell \times \omega = 0$ , are the six points  $\pm \ell_0 e_i$ : steady rotation about a principal axis. The middle level,  $2H = \ell_0^2/I_2$ , is

$$\ell_1^2 \left( \frac{1}{I_1} - \frac{1}{I_2} \right) = \ell_3^2 \left( \frac{1}{I_2} - \frac{1}{I_3} \right), \quad (8.19)$$

two planes  $\ell_3 = \pm k \ell_1$  through the  $e_2$  axis, cutting the sphere in two great circles that cross at  $\pm \ell_0 e_2$ : the *separatrix*.

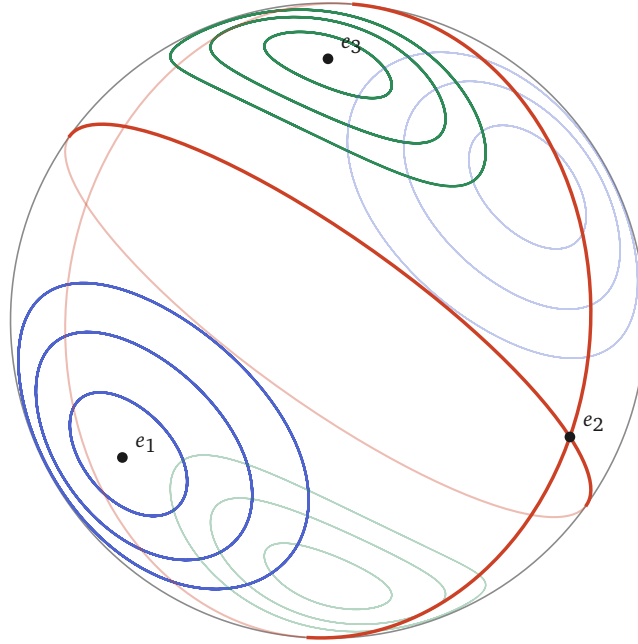


Figure 8.6: The free Euler top in the body frame,  $I = (1, 2, 3)$ ,  $|\ell| = 1$ . The closed loops round the axes of smallest and largest moment,  $\pm e_1$  (blue) and  $\pm e_3$  (green), are orbits of (8.16), computed by integrating it (`scripts/figures/euler_sphere.py`, which checks that  $|\ell|$  and  $H$  are conserved along each). The separatrix (red) is drawn from (8.19): two great circles, made of the equilibria  $\pm e_2$  and four orbits running between them. The far hemisphere is faint.

**Theorem 8.5 (The tennis-racket theorem).** *Let  $I_1 < I_2 < I_3$  and  $\ell_0 > 0$ . Steady rotation about the axis of smallest or of largest moment is stable: for every  $\varepsilon > 0$  there is  $\delta > 0$  such that  $|\ell(0) - \ell_0 e_1| < \delta$  implies  $|\ell(t) - \ell_0 e_1| < \varepsilon$  for all  $t$ , and the same for  $e_3$ . Steady rotation about the middle axis is unstable: there are solutions starting arbitrarily close to  $\ell_0 e_2$  that tend to  $-\ell_0 e_2$ .*

At the board the proof is Figure 8.6: near  $\pm e_1$  and  $\pm e_3$  the conserved quantities confine  $\ell$  to small loops, while every neighbourhood of  $\pm e_2$  meets the separatrix, which runs to the opposite pole. The reading turns both halves into estimates.

*Proof.* *Stability at  $e_1$ .* The function

$$G = \frac{|\ell|^2}{I_1} - 2H = \ell_2^2 \left( \frac{1}{I_1} - \frac{1}{I_2} \right) + \ell_3^2 \left( \frac{1}{I_1} - \frac{1}{I_3} \right) \geq c (\ell_2^2 + \ell_3^2), \quad c = \frac{1}{I_1} - \frac{1}{I_2} > 0,$$

is conserved and vanishes at  $\ell_0 e_1$ . So  $\ell_2^2 + \ell_3^2 \leq G(0)/c$  for all  $t$ , and  $G(0)$  is small when  $\ell(0)$  is near  $\ell_0 e_1$ . Since  $|\ell|$  is conserved,  $\ell_1^2 = |\ell|^2 - \ell_2^2 - \ell_3^2$  stays close to  $\ell_0^2$ ;  $\ell_1$  cannot pass through 0, so it keeps its sign and stays close to  $\ell_0$ . At  $e_3$  use  $2H - |\ell|^2/I_3$  instead.

*Instability at  $e_2$ .* The components of  $\ell \times \omega$  are  $\ell_2 \ell_3(1/I_3 - 1/I_2)$ ,  $\ell_3 \ell_1(1/I_1 - 1/I_3)$  and  $\ell_1 \ell_2(1/I_2 - 1/I_1)$ ; with distinct moments they all vanish only when two components of  $\ell$  do, so the equilibria are the points  $\pm \ell_0 e_i$ . Each open arc of the separatrix between  $\ell_0 e_2$  and  $-\ell_0 e_2$  contains no equilibrium, so a solution started on it moves along it with nowhere-vanishing speed, so monotonically; being bounded, it converges as  $t \rightarrow \pm\infty$ , and the limits can only be equilibria. So each arc runs from one end to the other. It remains to see that some arc leaves  $\ell_0 e_2$ . The map  $(\ell_1, \ell_2, \ell_3) \mapsto (-\ell_1, \ell_2, \ell_3)$  combined with  $t \mapsto -t$  takes solutions of (8.16) to solutions (check it on the three components, each of which is a product of the other two  $\ell$ 's), fixes  $\ell_0 e_2$ , exchanges the two planes of (8.19) and reverses time. So if the arc in one plane runs into  $\ell_0 e_2$ , its mirror image runs out of it, towards  $-\ell_0 e_2$ . Its points can be taken as close to  $\ell_0 e_2$  as we like. ■

Throw a tennis racket or a phone spinning about its middle axis: it flips, and off the separatrix it flips back periodically, because a nearby orbit follows a loop that runs close to the separatrix, round to near  $-\ell_0 e_2$  and back (Figure 8.7).

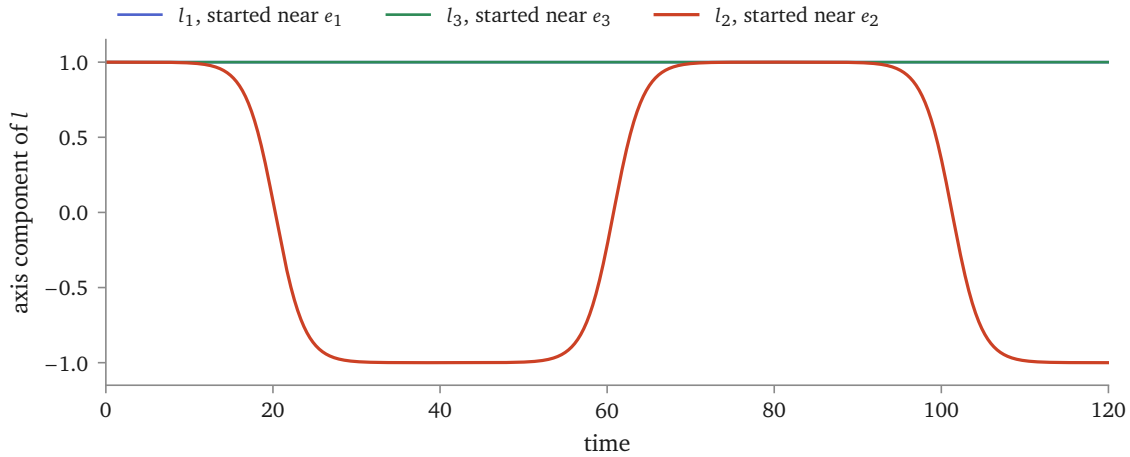


Figure 8.7: The tennis-racket theorem, numerically:  $I = (1, 2, 3)$ ,  $|\ell| = 1$ , each run started 0.01 radians from a principal axis. The axis components  $\ell_1$  (started near  $e_1$ ) and  $\ell_3$  (near  $e_3$ ) stay at 1 to within  $2 \times 10^{-4}$  and their curves lie on top of each other;  $\ell_2$  (near  $e_2$ ) flips between +1 and -1 over and over (scripts/figures/tennis\_racket.py, which asserts both).

### Poinsot's construction

The same two integrals give Poinsot's picture of the motion *in space*. In the body, the angular velocity lies on the *energy ellipsoid*  $E = \{v : v \cdot Iv = 2H\}$ , since  $\omega \cdot I\omega = \omega \cdot \ell = 2H$ . The normal to  $E$  at  $\omega$  is the gradient  $2I\omega = 2\ell$ , and the tangent plane there lies at distance  $\omega \cdot \ell/|\ell| = 2H/|\ell|$  from the centre. In space  $\ell$  is a fixed vector and  $2H/|\ell|$  a fixed number, so this tangent plane is *the same plane at all times*, the *invariable plane*. The point of contact is  $\omega$ , which lies on the instantaneous axis of rotation and so has zero velocity. Hence:

**Proposition 8.6 (Poincot).** *The energy ellipsoid, carried by the body, rolls without slipping on a fixed plane perpendicular to  $\ell$  at distance  $2H/|\ell|$  from the centre.*

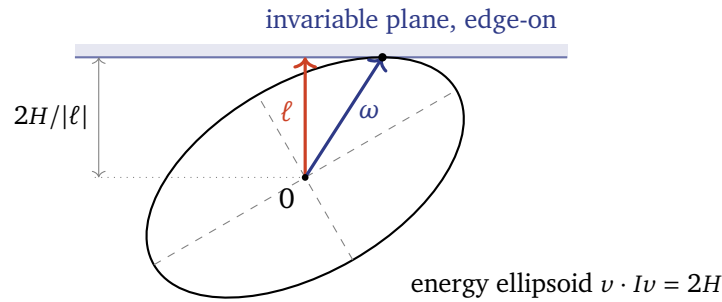


Figure 8.8: Poincot's construction.  $\ell$  is fixed in space; the energy ellipsoid, fixed in the body, touches the invariable plane at  $\omega$ , the one point of the ellipsoid with zero velocity, and so rolls on it without slipping. Drawn in the plane containing  $\ell$  and  $\omega$ .

**Remark 8.7 (The other tops).** The symmetric heavy top of Lagrange and the Kowalevski top also have Lax pairs, both with a spectral parameter of the kind § 8.11 introduces. *cited:* Audin, *Spinning Tops*, ch. II §2 and ch. III §2. We do not use them.

Reading

## 8.11 Gauge, and a spectral parameter

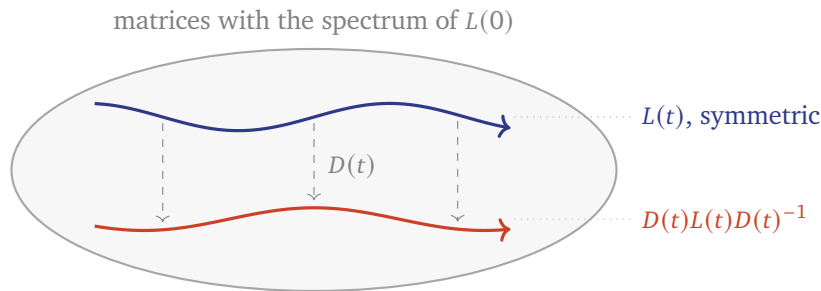


Figure 8.9: Gauge freedom, schematically. Both curves lie in the set of matrices with a given spectrum; each is the image of the same Toda trajectory under a different Lax matrix, and the gauge transformation  $D(t)$  (dashed) carries one to the other at each instant.

The warning to carry forward: the Poisson brackets of the *entries* of  $L$ , which [Lecture 9](#) will make the key to involution, change under a gauge transformation even though the spectrum does not. And the site-matrix section of this lecture uses a gauge transformation to make the periodic chain's Lax representation *local*, built site by site; [Lecture 9](#) uses the resulting site matrices to prove involution.

The Euler top left a problem: for a body whose moments are not all equal, its Lax matrix  $\hat{\ell}$  conserves  $|\ell|^2$  and misses  $H$  (the insufficiency discussed in § 8.3). Manakov's repair (1976)

is to let the Lax matrix depend on an extra complex number  $\lambda$ , the *spectral parameter*.

Put  $A = \text{diag}(A_1, A_2, A_3)$  with  $A_i = \frac{1}{2}(I_1 + I_2 + I_3) - I_i$ , so that  $A_2 + A_3 = I_1$  and cyclically. Since  $(A\Omega + \Omega A)_{jk} = (A_j + A_k)\Omega_{jk}$ , this gives  $\hat{\ell} = A\Omega + \Omega A$ . Now set

$$L(\lambda) = \hat{\ell} + \lambda A^2, \quad M(\lambda) = -(\Omega + \lambda A). \quad (8.20)$$

Then  $\dot{L}(\lambda) = [M(\lambda), L(\lambda)]$  for every  $\lambda$ : the  $\lambda^0$  term is (8.17), the  $\lambda^2$  term is  $[A, A^2] = 0$ , and the  $\lambda^1$  term is  $-\left[\Omega, A^2\right] - [A, A\Omega + \Omega A]$ , which vanishes because  $[A, A\Omega + \Omega A] = A^2\Omega - \Omega A^2$ .

So Theorem 8.2 applies at each  $\lambda$  separately, and the characteristic polynomial of  $L(\lambda)$  is conserved as a polynomial in two variables. It is

$$\det(\eta - L(\lambda)) = \det(\eta - \lambda A^2) + |\ell|^2 \eta - \lambda \sum_i A_i^2 \ell_i^2, \quad (8.21)$$

where the first term involves only the constants  $A_i$ , and

$$\sum_i A_i^2 \ell_i^2 = 2I_1 I_2 I_3 H + \alpha |\ell|^2, \quad \alpha = \frac{1}{4}(I_1^2 + I_2^2 + I_3^2 - 2I_1 I_2 - 2I_2 I_3 - 2I_3 I_1).$$

The coefficient of  $\lambda \eta^0$  contains  $H$ : the integral that  $\hat{\ell}$  missed is a spectral invariant of  $L(\lambda)$ . (For a spherical body  $A$  is a multiple of  $\mathbb{1}$ ,  $\lambda A^2$  only shifts the spectrum, and there was nothing to recover.) All three identities are checked symbolically in `scripts/check-17-algebra.py`.

This is the first thing a spectral parameter is for: *one matrix family carries more invariants than one matrix*, since every coefficient of every power of  $\lambda$  in  $\text{tr } L(\lambda)^k$  is conserved. There are two more, and they are the reason Part I keeps  $\lambda$  from here on.

- *A curve.* The conserved polynomial  $\det(\eta - L(\lambda))$  defines the *spectral curve*  $\Gamma = \{(\lambda, \eta) \in \mathbb{C}^2 : \det(\eta - L(\lambda)) = 0\}$ , which the flow leaves unchanged; its coefficients are the integrals. For the Toda ring, Lecture 9's reading forms a different but related curve,  $\det(z - T(\lambda)) = 0$  from the monodromy  $T$ , identifies its finite gaps from  $|\text{tr } T(\lambda)| > 2$ , and cites their relation to the action variables. (Here  $\eta$  is local notation.)
- *A bracket.* For the Toda ring, Lecture 9 checks a bracket that compares the site matrices  $L_n(\lambda)$  and  $L_n(\mu)$  at two *different* values of the parameter, composes it for their product round the ring, and proves that the traces at  $\lambda$  and  $\mu$  commute. That pays this lecture's IOU for the ring. Manakov's matrix is a separate example of what a spectral parameter can encode.

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Reading

## 8.12 Time evolution, zero curvature and plaquette holonomy

### Time evolution, and zero curvature

The site matrix transports  $\Phi$  in space. Transport in time is a second matrix at each site,

$$V_n(\lambda) = \begin{pmatrix} -\lambda/2 & -e^{-q_n} \\ e^{q_{n-1}} & \lambda/2 \end{pmatrix}, \quad (8.22)$$

which follows from  $\dot{\psi} = M\psi$ , the recursion and the same gauge (we omit that derivation; what matters is the next statement, checked directly).

**Proposition 8.8** (Discrete zero curvature). *For every  $\lambda$ ,*

$$\dot{L}_n(\lambda) = V_{n+1}(\lambda) L_n(\lambda) - L_n(\lambda) V_n(\lambda) \quad \text{for all } n \quad (8.23)$$

*holds if and only if  $\dot{q}_n = p_n$  and  $\dot{p}_n = e^{q_{n-1}-q_n} - e^{q_n-q_{n+1}}$ : the equations of motion of the ring.*

*Proof.* Multiply out:

$$V_{n+1}L_n - L_nV_n = \begin{pmatrix} e^{q_n-q_{n+1}} - e^{q_{n-1}-q_n} & -p_n e^{-q_n} \\ -p_n e^{q_n} & 0 \end{pmatrix}, \quad \dot{L}_n = \begin{pmatrix} -\dot{p}_n & -\dot{q}_n e^{-q_n} \\ -\dot{q}_n e^{q_n} & 0 \end{pmatrix}.$$

Every  $\lambda$  has cancelled on the right. The off-diagonal entries agree iff  $\dot{q}_n = p_n$ , the diagonal ones iff  $\dot{p}_n$  has the stated value. ■

Why “curvature”? Draw the lattice of sites and times (Figure 8.10). Put  $L_n$  on each horizontal link, as transport from site  $n$  to  $n+1$ , and  $\mathbb{1} + V_n dt$  on each vertical link, as transport from  $t$  to  $t+dt$ . Go round one plaquette, out along one pair of sides and back along the other:

$$(\mathbb{1} + V_n dt)^{-1} L_n(t+dt)^{-1} (\mathbb{1} + V_{n+1} dt) L_n(t) = \mathbb{1} + dt L_n^{-1} (V_{n+1} L_n - L_n V_n - \dot{L}_n) + O(dt^2). \quad (8.24)$$

This is the *holonomy* of the plaquette. In a flat geometry transport is path-independent and every small loop has trivial holonomy; so (8.23) says *the connection  $(L_n, V_n)$  on the space-time lattice is flat*. It is the discrete form of the zero-curvature condition

$$\partial_t U - \partial_x V + [U, V] = 0, \quad (8.25)$$

to which (8.23) reduces if  $L_n = \mathbb{1} + \Delta U(x)$ ,  $V_{n+1} = V + \Delta \partial_x V$  and terms of order  $\Delta^2$  are dropped. That equation, which integrates KdV and sine-Gordon, belongs to the sequel on field theory; it appears here once. The site-by-site transport, and its product round the ring, is the architecture [Lecture 14](#) quantises, with the Sklyanin bracket becoming the *RTT* relation.

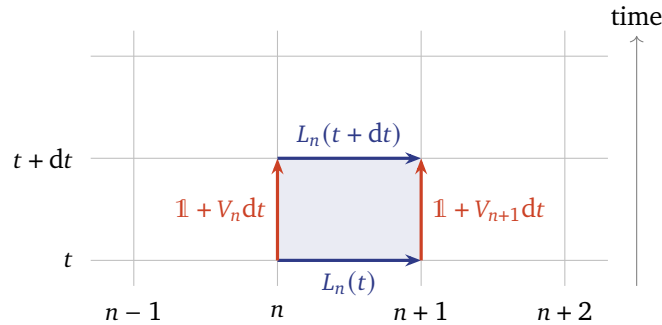


Figure 8.10: A plaquette of the space-time lattice. The two paths from lower left to upper right give  $(\mathbb{1} + V_{n+1} dt) L_n(t)$  and  $L_n(t+dt)(\mathbb{1} + V_n dt)$ ; zero curvature (8.23) says they agree to first order in  $dt$ .

### Summary.

- *Lax pair* (Definition 8.1):  $\dot{L} = [M, L]$  along the flow. Then the spectrum of  $L$  is conserved and the  $\text{tr } L^k$  are first integrals (Theorem 8.2); only  $N$  of them are independent.

- *Conserved is not in involution*: nothing in the Lax equation makes the  $\text{tr } L^k$  commute (§ 8.3).
- *Toda*: in Flaschka's variables  $L$  is the symmetric tridiagonal matrix fixed by  $\text{tr } L^2 = 2H$  and locality, and  $M$  is solved for and unique (Proposition 8.3); the first integrals are (8.8).
- *The ring*:  $a_N$  in the corners and  $a_1 \cdots a_N = 1$ ; the  $\text{tr } L^k$  are conserved for  $N \geq 3$ .
- *The site matrix*: in a gauge that makes it local, the eigenvalue problem is the transport  $\Phi_{n+1} = L_n(\lambda)\Phi_n$ , with  $\det L_n(\lambda) = 1$  (8.15).
- *Next*: Lecture 9 multiplies the site matrices round the ring and proves that the integrals commute.

**Tutorial.** Exercise Sheet 8.

## References

Babelon, Bernard and Talon, *Introduction to Classical Integrable Systems*, ch. 3 (Lax pairs) and ch. 2 on Toda; Toda, *Theory of Nonlinear Lattices*, §§2.10–2.11 (Flaschka's variables and the periodic chain); H. Flaschka, Phys. Rev. B 9 (1974) 1924; M. Hénon, Phys. Rev. B 9 (1974) 1921; M. Audin, *Spinning Tops* (the Euler top, in the reading).